Raspberry Pi Pico

P1
rpi pico 1
2648-SC0915CT-ND
tht

+3.3V
R1 220
3V1 green
GND

R10 220k
GND

+BATT
+3.3V
GND
J1
5V

NT5 NetTie_2
NT6 NetTie_2
NT7 NetTie_2
NT8 NetTie_2

bt1
bt2
bt3
bt4

1.8–5.5v
5V
D1 D_Schottky
+3.3V

VBUS
VSYS
3V3_OUT
ADC_VREF
3V3_EN
RUN
GP0
GP1
GP2
GP3
GP4
GP5
GP6
GP7
GP8
GP9
GP10
GP11
GP12
GP13
GP14
GP15
GP16
GP17
GP18
GP19
GP20
GP21
GP22
GP26
GP27
GP28
AGND
GND
GND_1
GND_2
GND_3
GND_4
GND_5
GND_6

R2 220
D13 blue
GND

sda
scl
right_1
right_2
right_en
left_2
left_en
left_1

+3.3V
J3 joystick
GND

+3.3V
J4 speed knob
GND

GND

left motor

U1
DRV8245HQPWPRQ

LVL6, in independent mode that means faults are retried

left_2
left_1
+BATT
GND
C1 0.1uF

SR
DIAG
PH/IN2
EN/IN1
DRVOFF
VM
OUT1
OUT2
GND
EPAD
ITRIP
MODE
NFAULT
IPROPI
NSLEEP

8.2k +-10% LVL2 INDEPENDENT
8.2k R7
GND
left_ipropi
left_en
+BATT

HSL1 Heatsink

J100 0197054001
J101 0197054001
optionally replace J7 to be weelye compatible

ML+
ML-
J7 1017491

right motor

U2
DRV8245HQPWPRQ

right_1
right_2
+BATT
GND
C2 0.1uF

8.2k +-10% LVL2 INDEPENDENT
8.2k R6
GND
right_ipropi
right_en
+BATT

HSR1 Heatsink

MR-
MR+
J8 1017491

optionally replace J8 to make the PCB weelye compatible
J104 2159151002

ADC (for measuring battery voltage and motor current)

U4
TLA2024IRUGR

+3.3V
NT1 NetTie_3
GND
TP0 TestPoint
ADDR
AIN0
AIN1
AIN2
AIN3
NC
VDD
SCL
SDA
GND
scl
sda
right_ipropi
left_ipropi
C4 0.1uF

+BATT
R13 100k
R14 13.7k
GND

TP1 TestPoint
left_ipropi
TP2 TestPoint
right_ipropi
R11 604
R12 604
GND

+3.3V
R20 EXB34V182JV
pullup resistors
1.8k
sda
scl

qwiic I2C plug
GND
+3.3V
sda
scl
J2 Conn_01x04
MP

weelye compatible header
J11
+BATT
switch
D2 D_Schottky
bat-
GND
Conn_01x10_Pin

3.5mm audio jacks for if you want to connect buttons as an alternative to the joystick

B1 ASJ-200-B-HT
B2 ASJ-200-B-HT
B3 ASJ-200-B-HT
B4 ASJ-200-B-HT
bt1
bt2
bt3
bt4
GND

battery power circuit

+BATT
J102 0197054001
J103 0197054001
optionally replace J5 to be weelye compatible
J5 TB005-762-02BE
bat-
R3 8.2k
C5 33uf
SW1 screw terminal
GND
18V zener D3
R4 100k
switch
J6
PSMN1R2-25YLDX NMOS Q1
reverse polarity protection
PSMN1R2-25YLDX Q2
on/off switch

3mm diameter bolt holes
3.35in x 2.2in
85.09mm x 55.88mm
H1 M3
H2 M3
H3 M3
H4 M3

alternative voltage regulator

Instead of the voltage regulator circuit, you could get a single THT module that outputs 5V.
Make sure the pinout matches.
Make sure the min and max input voltage are compatible with your battery.

U5
D78B05T, RBT05W24S05, pololu5593
+BATT
IN
GND
OUT
5V
GND

voltage regulator circuit https://webench.ti.com/appinfo/webench/scripts/SDP.cgi?ID=F7F7D983B50C08E2

+BATT
R5 590k
U3
TPS54202HDDCR
EN
FB
VIN
BOOT
SW
GND
C6 0.1uF
L1 12uH
C7 56pF
R8 100k
R9 13.7k
C8 47uF
C3 10uF
5V
GND

https://github.com/gobabygocarswithjoysticks gobabygocarswithjoysticks@gmail.com

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